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Sandia develops math techniques to improve computational efficiency in quantum chemistry

DOE/SANDIA NATIONAL LABORATORIES

LIVERMORE, Calif. - Researchers at Sandia National Laboratories have developed new mathematical techniques to advance the study of molecules at the quantum level.

Mathematical and algorithmic developments along these lines are necessary for enabling the detailed study of complex hydrocarbon molecules that are relevant in engine combustion.

Existing methods to approximate potential energy functions at the quantum scale need too much computer power and are thus limited to small molecules. Sandia researchers say their technique will speed up quantum mechanical computations and improve predictions made by theoretical

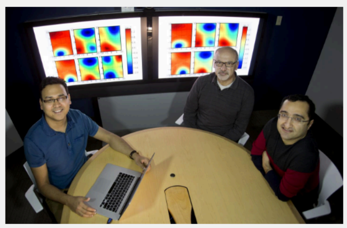

IMAGE: SANDIA NATIONAL LABORATORIES RESEARCHERS PRASHANT RAI, LEFT, HABIB NAJM, CENTER, AND KHACHIK SARGSYAN DISCUSS MATHEMATICAL TECHNIQUES USED TO STUDY THE BEHAVIOR OF LARGE MOLECULES AT QUANTUM SCALE. view more ›

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Plants at the pump

Sandia seeks toughest strains of algae for biofuel production as part of multilab project

By Jules Bernstein

Regular, unleaded, or algae?

That's a choice drivers could make at the pump one day. But for algal biofuels to compete with petroleum, farming algae has to become less expensive. Toward that goal, Sandia is testing strains of algae for resistance to a host of predators and diseases, and learning to detect when an algae pond is about to crash.

These experiments are part of the new, $6 million Development of Integrated Screening, Cultivar Optimization and Validation Research (DISCOVR) project, whose goal is to determine which algae strains are the toughest and most commercially viable.

ALGAE ABUSE – Sandia biochemist Carolyn Fisher (8623) examines a beaker full of microscopic algae eaters called rotifers being grown for the DISCOVR project. (Photo by Dino Vournas)

DOE's Office of Energy Efficiency and Renewable Energy sponsors the project, and Sandia's partners are Los Alamos (LANL) and Pacific Northwest (PNNL) national laboratories, the National Renewable Energy Laboratory (NREL), and Arizona State University (ASU).

Algae is a desirable biofuel source because it doesn't compete with other plants that serve as sources of food. However, an estimated 30 percent of current production on algae farms is lost each year due to pond crashes.

The national labs, and Sandia in particular with its expertise on algae predators, are uniquely suited for this research. Sandia is using its 1,000-liter indoor algae raceway facility, also called a "crash lab," to perform experiments that industrial groups will not do because they can't afford to contaminate their ponds.

"We use organisms and agents that many of my industrial partners do not allow on their sites," says biologist Todd Lane (8623), Sandia's project lead. "They cannot culture these creatures in their own facilities. It's too much of a risk."

Assembling 'a diverse panel of nasty things'

Todd and his team are cultivating what he calls "a diverse panel of nasty things" to learn which type of parasitic fungus, bacterium, or disease kills various strains of algae the quickest.

Perhaps the most threatening member of the predatory panel is a rotifer, a microscopic organism capable of eating 200 algal cells per minute. Rotifer infection can take a 132,000-gallon commercial pond from a healthy green to collapse within 48 hours. "They are basically tiny vacuums for algae," says Todd.

This algae abuse will begin in March, using a strain called nanochloropsis as a baseline for survival statistics. Ironically, nanochloropsis is an industrial algae strain typically grown for feeding rotifers on fish farms. The team will conduct these tests on 15 algae strains in the crash lab under light and temperature conditions that mimic a variety of outdoor environments.

The ability to simulate different climates further enhances the team's ability to validate certain strains. However, even the strongest algae are susceptible to infections, so detecting infections before ponds crash is key.

Reflected light can indicate pathogen presence

For early crash detection, Jeri Timlin (8631), an analytical chemist on Sandia's DISCOVR team, is using a technique called spectro-radiometric monitoring to watch the ponds for subtle changes in reflected light that indicate the presence of pathogens or predators. Sandia researcher Tom Reichardt (8128), a project adviser, developed the technique. Most objects reflect light in different wavelengths, which result in the perception of color. This technique can detect subtle color changes as well as other physical and chemical properties of the algae, making it possible to determine the pond's density and overall health.

A major advantage of the Sandia monitoring method is that it provides real-time measurements without lab analysis. Previously, scientists had to do the time-consuming task of taking a sample from a pond back to a lab to measure it. Spectro-radiometric monitoring takes precise pond measurements every five minutes, without physical contact with the pond itself. And, no human is needed to make the measurements.

Algae pipeline to economic competitiveness

While Sandia monitors ponds and evaluates resistance to diseases, PNNL will quantify the biomass production rate of 10 strains of algae that they grow in a variety of simulated environmental conditions. NREL then will perform compositional analysis on the same strains, seeking those best suited for fuel production.

TINY PREDATOR – Microscope photo of a rotifer grown at Sandia National Laboratories shows a belly full of algae. (Image provided by Sandia)

Later phases of the three-year project will involve partners in this "algae pipeline," increasing pond culture stability and evaluating the potential for generating products other than fuel, such as chemicals used for industrial purposes. LANL will take the strongest strains and work to improve them further, aiming for faster growth rates and higher tolerance for a range of environmental conditions. Finally, ASU will validate the national labs' research in outdoor test beds.

By determining the most resilient algae strains and best practices for algae farming, Sandia and its partner labs may one day enable farmers to produce enough algae to make biofuels a real competitor at the pump.

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Reversing the curse

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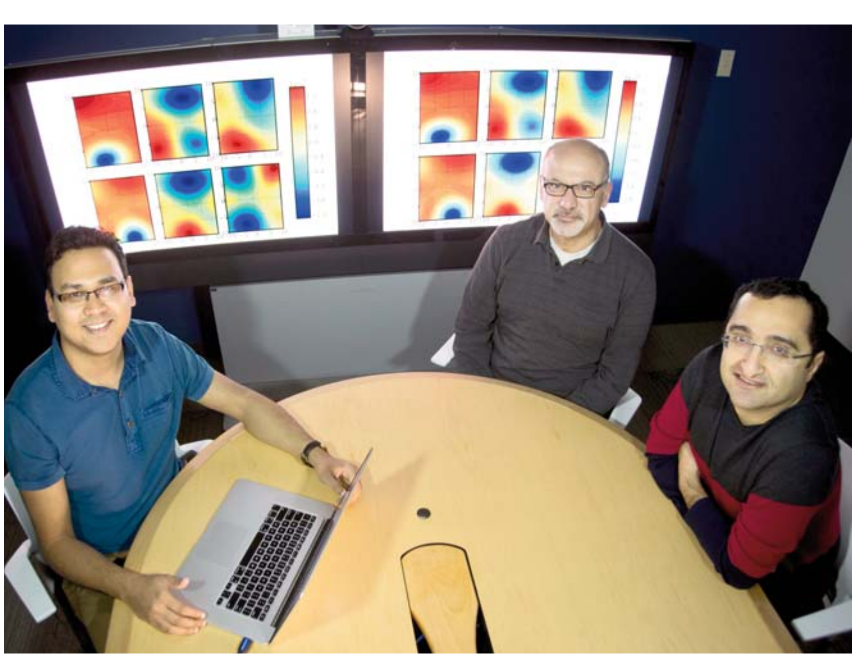


SANDIA RESEARCHERS Prashant Rai, Khachik Sargsyan, and Habib Najm.

By Michael Padilla

Sandia researchers have developed new mathematical techniques that can be used to study the behavior of large molecules at quantum scale. This technology helps reduce computation time and improves the predictive capability of theoretical chemistry models.

Sandia postdoctoral researcher Prashant Rai, working with Khachik Sargsyan and Habib Najm (all 8351) at the Combustion Research Facility (CRF), in collaboration with quantum chemists So Hirata and Matthew Hermes from the University of Illinois at Urbana-Champaign, developed computationally efficient methods to approximate potential energy surfaces of molecules by probing them at fewer configurations as compared to classical methods. Understanding potential energy surfaces, key elements in virtually all calculations in quantum dynamics, is required to accurately estimate the energy and frequency of vibrational modes of molecules.

The initial, promising results of this research will be published in *Molecular Physics* in an article titled "Low-rank canonical-tensor decomposition of potential energy surfaces: Application to grid-based diagrammatic vibrational Green's function theory."

"Approximating potential energy surfaces of bigger molecules is an extremely challenging task due to the exponential increase in information required to describe them with each additional atom in the system," Prashant says. "In mathematics, it is termed the Curse of Dimensionality."

The key to beating the Curse of Dimensionality is to exploit the characteristics of the specific structure of the potential energy surfaces. Knowledge of this structure can then be used to approximate the requisite high dimensional functions.

"We make use of the fact that although potential energy surfaces can be high-dimensional, they can be well-approximated as a small sum of products of one-dimensional functions. This is known as the low rank structure where the rank of the potential energy surface is the number of terms in the sum," Prashant says. "Such an assumption on structure is quite general and has also been used in similar problems in other fields. Mathematically, the intuition of low rank approximation techniques comes from multilinear algebra where the function is interpreted as a tensor and is decomposed using standard tensor decomposition techniques."

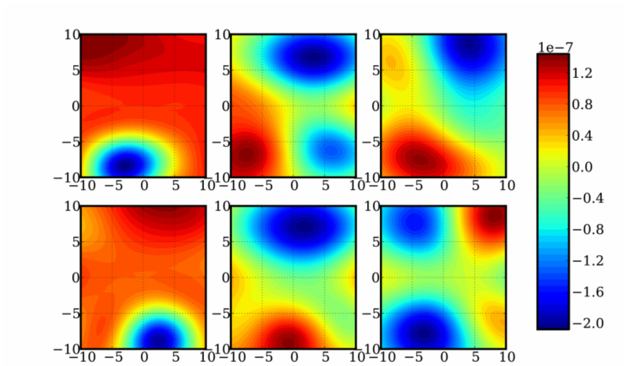
Energy and frequency corrections to classical results are formulated as integrals of these high-dimensional functions. Approximation in such a low rank format renders these functions easily integrable as it breaks the integration problem to the sum of products of one or two dimensional integrals for which standard integration methods apply.

This method has been initially applied and tested on small molecules such as water and formaldehyde. When compared to the classical Monte Carlo method, a standard workhorse for high dimensional integration problems, this approach improves the computational efficiency by orders of magnitude with better accuracy at the same time. Rai said the next step in this study is to further enhance the technique by challenging it with bigger molecules such as benzene.

Interdisciplinary studies such as these provide opportunities for cross pollination of ideas, thereby providing a new perspective on problems and their possible solutions. In the context of combustion research at the CRF, this work advances the computational capabilities available for the detailed study of complex hydrocarbon molecules relevant in engine combustion.

Team develops math techniques to improve computational efficiency in quantum chemistry

May 3, 2017, Sandia National Laboratories



A depiction of a random two-dimensional slices of a 12-dimensional function for determining energy and frequency corrections of a formaldehyde molecule. Credit: Sandia National Laboratories

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Existing methods to approximate potential energy functions at the quantum scale need too much computer power and are thus limited to small molecules. Sandia researchers say their technique will speed up quantum mechanical computations and improve predictions made by theoretical chemistry models. Given the computational speedup, these methods can potentially be applied to bigger molecules.

Sandia postdoctoral researcher Prashant Rai worked with researchers Khachik Sargsyan and Habib Najm at Sandia's Combustion Research Facility and collaborated with quantum chemists So Hirata and Matthew Hermes at the University of Illinois at Urbana-Champaign. Computing energy at fewer geometric arrangements than normally required, the team developed computationally efficient methods to approximate potential energy surfaces.

A precise understanding of potential energy surfaces, key elements in virtually all calculations of quantum dynamics, is required to accurately

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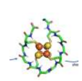


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